

AHMAD MARAQA

AI Research Engineer | AI/ML · LLM Agents · Computer Vision · AI Automation

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[Portfolio](#) · [LinkedIn](#) · [GitHub](#)

PROFILE

AI Research Engineer with professional experience across AI/ML, large language models, computer vision and AI automation. I train and evaluate machine learning models on real, noisy data — classification, regression, ensembles and neural networks — and build LLM-based systems on top of RAG, embeddings, vector databases and tool-calling agents. I also design and run the automation that connects those systems to databases, APIs and messaging platforms in production, using Python and n8n.

CORE SKILLS

AI & Machine Learning Classification · Regression · Binary & Multi-Class · Multi-Label Classification · Ensemble Methods · Model Evaluation · Deep Learning · Neural Networks · CNNs · Data Augmentation

LLMs & Agents LLMs · Retrieval-Augmented Generation (RAG) · Embeddings · Vector Databases (Pinecone, FAISS) · Prompt Engineering · Tool Calling · Structured Output · Agent Memory · Local & Open-Source LLMs

Computer Vision Object Detection · Image Classification · YOLO / YOLOv8 · OpenCV · Ultralytics · CVAT · Data Annotation

Frameworks & Tools Python · TensorFlow · Keras · Scikit-learn · Pandas · NumPy · Ollama · LangChain · LlamaIndex · OpenRouter · Groq · Google Gemini · ElevenLabs · Claude Code

Automation & APIs n8n (incl. self-hosted) · MongoDB · WhatsApp Business Cloud API · Telegram Bot API · Gmail, Google Calendar & Google Sheets APIs · REST APIs · Webhooks · Webhook Security · Rate-Limit Handling

Deployment & Ops Hostinger VPS · Cloudflare · Domains & DNS · Streamlit · GitHub Pages · Git & GitHub

Engineering Base IoT · BLE · RTLS · Microcontrollers · Verilog

PROFESSIONAL EXPERIENCE

AI Research & Integration Engineer — **Penguin Location Services** Jan 2025 – Present · Amman, Jordan

- Research and implement AI/ML solutions for real-world applications, including machine learning models for IoT and RTLS environments.
- Work with BLE and IoT devices and controllers in RTLS applications, applying machine learning models to noisy sensor data.
- Develop and experiment with LLM-based applications using Retrieval-Augmented Generation (RAG), embeddings, vector databases, and local/open-source LLMs.
- Apply prompt engineering techniques when developing and improving LLM-based applications.
- Use Ollama for working with local LLMs and AI application development.
- Build AI-powered assistants and integrations connecting language models with external services and tools.

AI Automation & Agents — Independent — **Self-taught** 2025 – Present · Amman, Jordan

- Develop and maintain production automation workflows using n8n, MongoDB, APIs, webhooks, and external integrations — 30+ workflows built to date.
- Integrate the WhatsApp Business Cloud API into automated workflows and AI-powered applications, including approved message templates, verified webhooks, batched sending within API rate limits, and timed jobs for media renewal and cleanup.
- Build tool-calling AI agents on the Telegram Bot API for task management, scheduling, document review and customer-facing support with escalation to a human.
- Integrate agents with Google Sheets, Google Calendar and Gmail, plus speech-to-text and text-to-speech, structured output parsing, and RAG over Pinecone.
- Self-host and maintain n8n on a Hostinger VPS, including deployment, credential management, HTTPS, persistence, domains, DNS and Cloudflare.

AI Research Engineer Intern — **Penguin Location Services** Oct 2024 – Jan 2025 · Amman, Jordan

- Worked with BLE and IoT devices in RTLS applications.
- Trained and evaluated machine learning models including KNN, Decision Trees and Random Forest.
- Implemented a custom probabilistic model for an RTLS-related application, keeping prediction confidence as part of the output.

AI & ML Engineer Intern — **Leading Point Software** Jul 2024 – Sep 2024 · Amman, Jordan

- Worked on data science, machine learning, deep learning and computer vision applications.
- Developed machine learning and computer vision solutions including image classification and object detection.
- Gained hands-on experience with AutoML tools.

SELECTED PROJECTS

Jordanian Currency Detection & Counting — YOLOv8 object detection over a dataset collected and augmented in-house, since no public dataset existed. Detects and counts notes and coins from a photo; deployed as a web app. [\[Live\]](#) [\[Code\]](#)

Fire Detection from IoT Sensors — Binary classification on ~62k tabular sensor readings across 13 features. Seven models trained and compared — logistic regression, decision tree, random forest, naive Bayes, gradient boosting, a voting ensemble and a Keras neural network — with all seven surfaced side by side in the app. [\[Live\]](#) [\[Code\]](#)

Rejecting Unknown Classes in Classification — Research into why a softmax classifier can never decline to answer. The same CNN was built twice, changing only the output layer and loss, so a multi-label formulation could score each class independently and reject inputs outside its training classes. [\[Notebook\]](#)

Business Messaging on the WhatsApp Cloud API — Production integration layer between a web application and the official WhatsApp Business Cloud API, with MongoDB as the source of truth, verified webhooks, rate-limit-aware batching, and scheduled maintenance jobs. Self-hosted on a VPS.

AI Agents & Assistants — A suite of production Telegram agents: a task manager that refuses to write incomplete records, voice-driven calendar and membership assistants using speech-to-text and text-to-speech, a CV review bot, and an email drafting agent scoped to drafts only.

Self-Updating RAG Knowledge Base — Retrieval-augmented generation over a document folder, indexing only files it has not seen before, with HuggingFace embeddings in Pinecone and retrieval exposed to the agent as a tool.

Dublejli — AI Video Dubbing System — End-to-end pipeline combining speech-to-text, translation and text-to-speech, with a synchronisation step aligning generated speech to the original video. Exhibited at SOFEX 2024 and the National Technology Parade (NTP) 2024.

EDUCATION

B.Sc. Computer Engineering — **University of Jordan** Oct 2020 – Sep 2024 · Amman, Jordan
GPA 3.77 / 4.00 — Excellent

LANGUAGES

Arabic — native • English — professional working proficiency